

Relationship Between Surface EMG Amplitude and Isometric Grip Force in the Dominant and Non-Dominant Forearm

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Abstract

Surface electromyography (sEMG) reflects neural drive to a muscle, but how well its amplitude tracks the force the muscle produces depends on the measurement and processing chain. This study recorded sEMG from the finger flexors of a single right-handed participant together with grip force from a hand dynamometer (BioRadio, 2000 Hz) during isometric handgrip contractions of varying intensity, including a maximal voluntary contraction (MVC), in both arms. EMG was demeaned, band-pass filtered (20–500 Hz), notch filtered (60 Hz), rectified and low-pass filtered (5 Hz) to obtain a linear envelope in MATLAB. The left arm produced a higher peak force than the right (225 N vs. 209 N) despite right-hand dominance. EMG onset preceded force onset by 0.295 s (left) and 0.378 s (right). Linear fits of mean force against mean EMG envelope had slopes of 1473.79 N/V (left) and 2559.15 N/V (right), but each fit rests on only two contractions, so the results are preliminary and do not establish linearity.

Keywords: *electromyography, grip strength, isometric contraction, EMG–force relationship, electromechanical delay, signal processing*

1. Introduction

Electromyography (EMG) measures the electrical activity produced by skeletal muscle during contraction. The EMG signal is the summation of motor unit action potentials, so its amplitude depends on the number of active muscle fibres and their firing rates. EMG amplitude therefore offers a window into neuromuscular activation and, indirectly, into the force a muscle generates.

Understanding how electrical activation translates into mechanical output is central to biomechanics, rehabilitation and motor-control research, and to the design of EMG-driven devices such as prostheses. In this experiment, finger flexor EMG was recorded simultaneously with grip force from a hand dynamometer. The objectives were to:

- characterise the relationship between EMG envelope amplitude and grip force;
- measure the electromechanical delay between EMG onset and force onset during maximal contractions; and
- compare both quantities between the dominant and non-dominant arms.

2. Methods

2.1 Apparatus and Electrode Placement

Data were acquired with a BioRadio system fitted with a hand dynamometer and surface EMG electrodes, and configured in BioCapture software with a 2000 Hz sampling rate, a differential EMG channel, and the dynamometer connected as the grip-strength sensor input. Three disposable snap electrodes were used per

arm: two placed along the belly of the finger flexor muscle and one reference electrode over the bony prominence of the elbow. The skin was abraded with prep gel and cleaned with alcohol wipes beforehand to reduce impedance and noise (Figure 1).



Figure 1. *Experimental setup showing sEMG electrode placement on the finger flexor and connection to the BioRadio and hand dynamometer.*

2.2 Protocol

The participant performed a series of isometric handgrip contractions at varying intensities, including one MVC. Each contraction was held for approximately three seconds, with rest intervals between trials to limit fatigue. The protocol was then repeated for the other arm so that neuromuscular activation could be compared between the dominant and non-dominant sides. EMG and grip force were recorded continuously throughout.

2.3 Signal Processing

All processing was performed in MATLAB, using the pipeline summarised in Table 1. Filters were applied forward and backward (MATLAB filtfilt) so that they introduce no phase shift.

Table 1. *EMG processing pipeline.*

Step	Operation	Parameters and purpose
1	Demmean	Subtract signal mean; removes constant offset and baseline drift
2	Band-pass filter	2nd-order Butterworth, 20–500 Hz; removes movement artefact and high-frequency noise
3	Notch filter	60 Hz IIR notch, 2 Hz bandwidth; removes mains interference
4	Full-wave rectification	Absolute value; amplitude reflects overall activation intensity
5	Low-pass filter	2nd-order Butterworth, 5 Hz cutoff; yields the linear envelope

2.4 EMG–Force Analysis

EMG envelopes and force were plotted on dual-axis graphs to compare activation and force over time. The electromechanical delay was defined as the time between EMG onset and force onset during the maximal

contraction. Within a ± 1 s window around peak force, onset was taken as the first sample at which each signal exceeded 10% of its maximum in that window.

For the force–EMG relationship, contractions were segmented wherever the envelope exceeded 30% of its overall maximum. The mean envelope amplitude and mean force were calculated for each segment, and a first-order polynomial (linear) fit was computed separately for each arm using MATLAB's polyfit function.

3. Results

3.1 EMG and Force Time Series

Figures 2 and 3 show the filtered EMG, its linear envelope and the grip force for the left and right arms. In both arms, bursts of EMG activity coincide with force peaks, and the largest EMG burst accompanies the MVC.

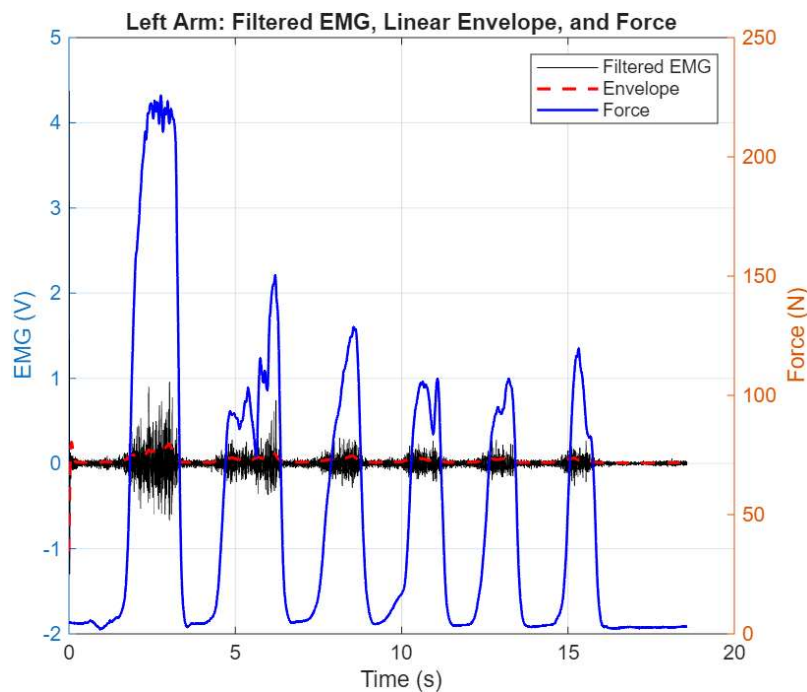


Figure 2. Filtered EMG, linear envelope and force against time, left arm.

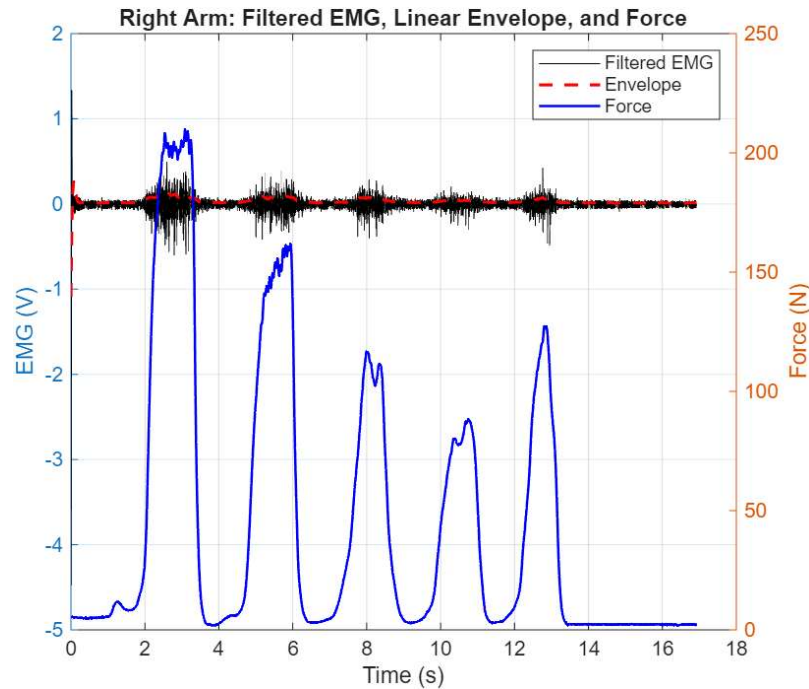


Figure 3. Filtered EMG, linear envelope and force against time, right arm.

3.2 Electromechanical Delay

Figures 4 and 5 show the onset of the maximal contraction in each arm. The calculated delay between EMG onset and force onset was 0.295 s for the left arm and 0.378 s for the right arm; this delay has been factored out in the plotted signals.

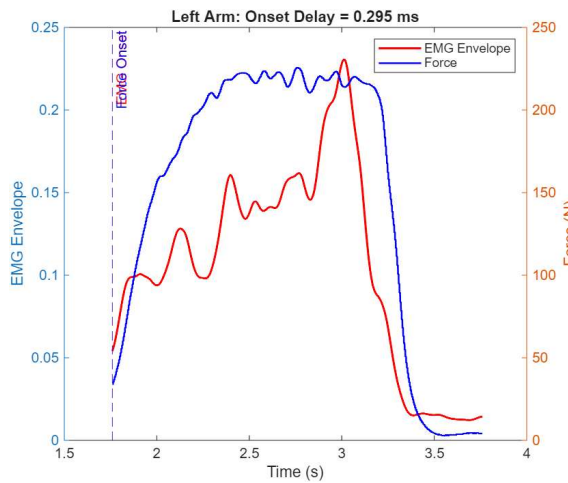


Figure 4. Maximum contraction onset, left arm (delay 0.295 s).

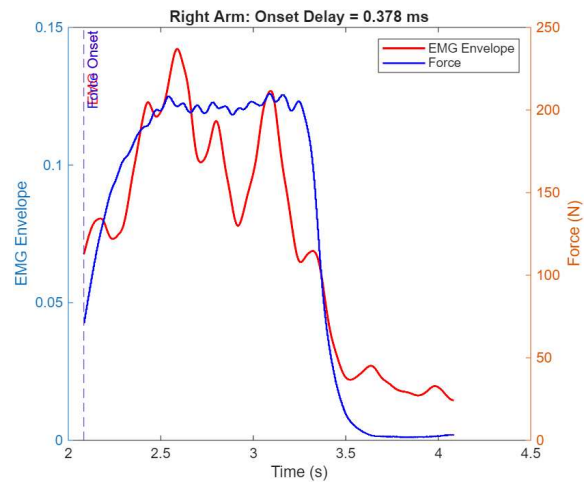


Figure 5. Maximum contraction onset, right arm (delay 0.378 s).

3.3 Force–EMG Relationship

Mean force is plotted against mean EMG envelope amplitude for each arm in Figure 6, with linear trendlines. Both arms show a positive slope; the right arm's slope is steeper. Table 2 summarises the main quantities. Note that each fit is based on two contractions.

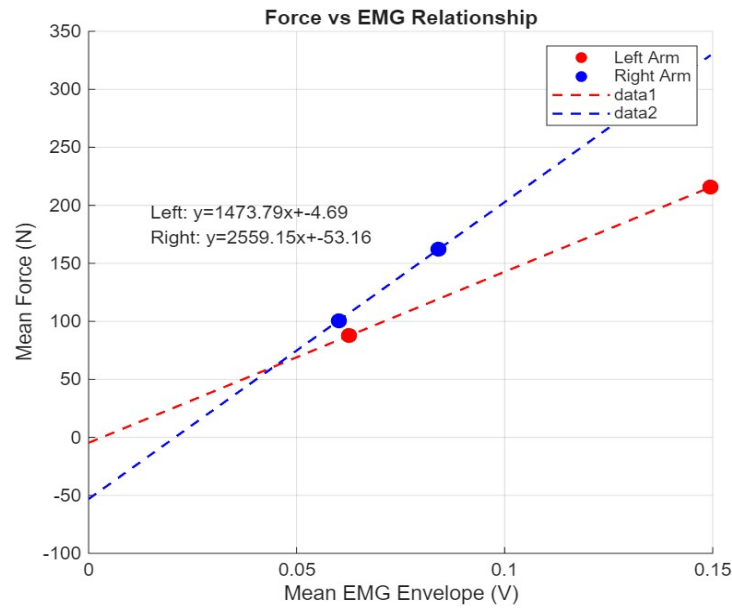


Figure 6. Mean force vs. mean EMG envelope for the left and right arms, with linear trendlines and equations.

Table 2. Summary of results by arm.

Arm	Peak force (N)	EMG–force delay (s)	Slope (N/V)	Intercept (N)	Points in fit
Left	225	0.295	1473.79	–4.69	2
Right (dominant)	209	0.378	2559.15	–53.16	2

4. Discussion

4.1 Maximal Force and Hand Dominance

The left arm produced a higher peak force (225 N) than the right (209 N), a difference of roughly 7%. Because the participant is right-handed, the opposite would normally be expected, as the dominant arm is used more often and tends to show greater neuromuscular efficiency. Several procedural factors could explain the result: differences in hand position on the dynamometer (wrist angle or finger placement) that alter mechanical leverage; fatigue if the right arm was tested after the left; and variation in electrode placement or signal interference. With one participant and no repeated trials, these explanations cannot be separated.

4.2 Force–EMG Relationship

Both arms show a positive relationship between mean EMG amplitude and mean force, consistent with the principle that greater neural activation recruits more muscle fibres and produces more force. The right-arm trendline is steeper (2559.15 N/V vs. 1473.79 N/V), which would suggest more force per unit of measured EMG in the dominant arm. However, this interpretation should be made cautiously: EMG amplitude scales with electrode position, skin impedance and crosstalk, so a difference in slope may reflect the recording rather than the muscle.

More importantly, each trendline is fitted to only two contractions (Figure 6). A line through two points fits perfectly by construction, so these fits cannot confirm that the relationship is linear or provide a goodness-

of-fit measure. The negative intercepts (-4.69 N and -53.16 N) are also non-physiological, since zero activation should produce roughly zero force, and are an artefact of extrapolating from two points. The trends should be read as directional evidence only.

4.3 Electromechanical Delay

The measured delays (0.295 s left, 0.378 s right) are considerably longer than the tens of milliseconds typically reported for electromechanical delay. This likely reflects the measurement approach rather than physiology alone: onset was detected with fixed 10% thresholds on a heavily smoothed (5 Hz) envelope, and grip force builds gradually against the dynamometer. The 83 ms difference between arms should therefore not be interpreted physiologically without a more precise onset-detection method and repeated trials.

4.4 Limitations

- Electrode placement was not standardised, and inconsistent placement directly changes EMG amplitude. Electrodes can also pick up signals from neighbouring muscles (crosstalk).
- Data come from a single participant, so the results do not generalise to a wider population.
- Small movements of electrodes or cables during contraction introduce noise that overlaps with the true muscle signal.
- Only two contractions per arm entered the force–EMG fits, which limits any conclusion about linearity.

4.5 Future Work

The experiment could be improved by adding participants and repeated trials, standardising electrode placement (for example, using anatomical landmarks) to reduce crosstalk, counterbalancing the order in which arms are tested to control for fatigue, and including more contractions across a wider range of force levels so that the force–EMG fit is properly constrained. Improved filtering and tighter EMG–force timing synchronisation would also give a more reliable estimate of electromechanical delay.

5. Conclusion

This study recorded finger flexor EMG and grip force during isometric contractions in both arms of one participant. EMG amplitude and force increased together in both arms, in line with the expected link between neural activation and force output. The left arm produced a higher peak force (225 N) than the right (209 N) despite right-hand dominance, most likely due to procedural factors such as hand positioning or fatigue, while the right arm showed a steeper force–EMG slope. EMG onset preceded force onset by 0.295 s (left) and 0.378 s (right). Given the single participant and the two-point fits, these findings are preliminary; more contractions, more participants and more standardised electrode placement would be needed to draw firmer conclusions.

Author Contributions

P. Khanvilkar: MATLAB analysis and results. F. A. Wahyudi: data analysis and background. C. J. Setiawan: discussion and summary.

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AI tools (Perplexity and ChatGPT) were used for guidance with the MATLAB code, including how to locate indices in MATLAB and whether the code met the lab guidelines.